

Mohammadsaeed Haghi

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EDUCATION

- **Ph.D. in Operations Research** | University of Southern California (USC) Aug 2024 -present
Under the Supervision of Prof. Phebe Vayanos
- **M.Sc. in Computer Science** | University of Southern California (USC) Jan 2024 -present
- **B.Sc. in Applied Mathematics** | Sharif University of Technology (SUT) Sep 2019 -Mar 2024
Class Rank : 2nd of 25

SELECTED AWARDS

- **Second Place in National Statistics Olympiad** Sep 2023
Nationwide highly selective contest covering advanced statistics, probability, and algorithmic programming.
- **Silver Medal in !Optimizer** | Sharif Optimization and Applications Laboratory(SOAL) July 2022
Competition organized annually by SOAL on optimization; in 2022, its theme was multi-manifold clustering.
[GitHub: Multi-Manifold Clustering](#)
- **Silver Medal in Iran Mathematical Olympiad** Sep 2018
Nationwide IMO style proof-based mathematics competition.

ACADEMIC EXPERIENCE

- **University of Southern California**
 - *Research Assistant Under the Supervision of Prof. Phebe Vayanos* Sep 2024 - present
Designed optimization-based methods for treatment assignment with capacity constraints under hidden confounders. Applied dual linear programming and KNN-based extensions; implemented in Python with Gurobi.
[GitHub: Policy Selection under Unobserved Confounder](#)
- **Sharif University of Technology**
 - *Research Assistant Under the Supervision of Prof. Javad Ebrahimi Broojeni* Jun 2023 - April 2024
Implemented a constrained optimization technique to approximate the shortest-path length distribution in graphs with respect to **KL-divergence**.
[GitHub: KL-divergence Minimizer](#)
 - *Research Assistant Under the Supervision of Prof. Mojtaba Tefagh* April 2023 - April 2024
Applied Hadamard and Clarkson-Woodruff sketches to reduce dimensions in constrained least squares problems. Worked on applications including radiotherapy beam angle selection.
[GitHub: Matrix Sketching](#)
 - *Class Project for Financial Mathematics class* Spring 2023
Implemented an algorithm to detect and correct arbitrage violations in predicted option prices. Applied no-arbitrage constraints to ensure consistency with financial theory and market structure.
[GitHub: Arbitrage Repair](#)

SKILLS

Programming Languages:	Python, Julia, MATLAB, Java
Optimization & Modeling:	CVXPY, Gurobi, MOSEK, Linear Programming, Duality Theory, Convex Optimization
Data Science & ML:	NumPy, Pandas, Scikit-learn, Statistical Learning, Stochastic Processes, Transformers
Finance & Economics:	Mathematical Finance, Game Theory, Econometrics, Decision Theory
Soft Skills:	Teaching Experience, Event & Competition Instruction, Analytical Thinking, Problem Solving